AKKA

Reliable AI for Every Industry
The Agentic Systems Platform

Akka vs. NVIDIA

A comparison for teams building agentic AI

NIM · NeMo · NeMo Agent toolkit · AI Enterprise · Blueprints | June 2026

NVIDIA's agentic stack serves and assembles AI; Akka governs and runs the agentic application above it. NIM, NeMo, the NeMo Agent toolkit, AI Enterprise, and Blueprints are an inference-and-microservice layer tied to NVIDIA GPUs and CUDA — software you self-host, assemble, and operate. The Akka Agentic AI Platform is the governed agentic application platform that runs on top of that serving layer and guarantees resilience and scalability, with a contractual availability SLA, durable agent state, and runtime governance.

None	99.9999%	4ms	Mar 2025
SELF-HOSTED UPTIME SLA	AKKA AVAILABILITY SLA	AKKA DURABLE-STATE READS	NEMO AGENT TOOLKIT 1.0

DIMENSION	NVIDIA (NIM / NEMO / NEMO AGENT TOOLKIT / AI ENTERPRISE / BLUEPRINTS)	AKKA
What it is	An inference + microservice-and-toolkit layer: containerized model serving, model customization, and an open-source agent library, packaged as self-hosted software	**The Akka Agentic AI Platform — a governed agentic application platform that guarantees resilience and scalability, running on one runtime**
Layer	The serving and assembly layer; you integrate and operate the application around it	**The application layer above model serving — agents, memory, streaming, APIs, governance pre-integrated**
Availability SLA	Support-response SLA only (4-hour initial response, 8x5); no uptime/availability SLA — you run the software in your own infrastructure	**99.9999% uptime — entire platform, backed by indemnities, Akka-operated**
RTO / RPO	Owned by the customer; no published RTO/RPO	**Sub-1-minute RTO; zero-byte RPO, active-active across regions**
Durable agent state / memory	Not provided as managed infrastructure; customer sources and operates a state/vector store	**Durable in-memory, 4ms reads / sub-10ms writes, replayable from the event journal**
Agent toolkit maturity	NeMo Agent toolkit: open-source library first released March 2025 (renamed from AIQ / Agent Intelligence toolkit); framework-agnostic glue for LangChain/LlamaIndex/CrewAI — not a hosted runtime	**Production runtime, since 2007, 100,000+ deployments**
Governance / EU AI Act	NeMo Guardrails provides content/topic rails; no inline runtime enforcement, immutable audit ledger, pre-deployment classification, or sealed posture	**Aspect-woven runtime enforcement + full pre-production governance**
Hardware coupling	Optimized for and dependent on NVIDIA GPUs and CUDA	**Runs on any hardware, any cloud, on-prem; no GPU/CUDA dependency for the application layer**
Cost model	AI Enterprise at $4,500 per GPU per year (or per-GPU-per-hour cloud) for the serving layer; you provision and operate everything above it	**Shared compute; up to 90% lower infrastructure for the same agentic workload, fixed annual fee**

Certifications	AI Enterprise production-branch support, monthly CVE patching, security notifications, SBOM/VEX	19 standards (SOC 2 II + public SOC 3, ISO 27001/42001, HIPAA, PCI DSS, GDPR, NIS2, DORA, EU AI Act, NIST AI RMF)

A Serving-and-Toolkit Layer, Not a Governed Application Platform

NVIDIA's agentic stack is five pieces of software the customer assembles and operates: **NIM** containers serve models, **NeMo** customizes them, the **NeMo Agent toolkit** wires agents together in code, **Blueprints** are reference samples to copy, and **NVIDIA AI Enterprise** is the licensed, security-maintained bundle that ships them. Each is real and well-built. Together they are a serving and assembly layer — not a governed agentic application platform with one runtime and one SLA.

Capability	NVIDIA stack	Akka
Model inference / serving	Yes — NIM (best in class)	Not provided (calls NIM or any endpoint)
Model fine-tuning / customization	Yes — NeMo Customizer	Not provided
Native agent runtime with durable execution	NeMo Agent toolkit is a library you host and operate	Built in
Durable agent memory as managed infrastructure	Customer sources and operates	Built in, 4ms / sub-10ms
Real-time streaming engine	Customer provisions separately	Built in, backpressured, petabyte-scale
HTTP / gRPC API layer	Customer builds	Built in
Inline governance / policy enforcement	Content rails (NeMo Guardrails) only	Inline, runtime-embedded
Pre-production governance	None	Classification, sign-offs, sealed posture
Operated runtime with an availability SLA	No — you self-host the software	Yes — Akka operates it at 99.9999%

NVIDIA serves and accelerates the model. Akka runs the application that uses it.

Availability: A Support SLA Is Not an Uptime SLA

NVIDIA AI Enterprise includes a **support-response SLA** — Business Standard Support with a 4-hour initial response time, 8am–5pm local business hours, plus software updates, maintenance branches, and security notifications. That is support for software you run yourself. Because you operate NIM, NeMo, and your agents in your own infrastructure, **the availability of the running system is owned by the customer** — there is no published uptime SLA, no RTO, and no RPO for the application. (NVIDIA's 99% Service Availability target applies only to its hosted DGX Cloud, a separate product.)

Metric	NVIDIA stack	Akka
Availability (uptime) SLA	None — customer-owned	99.9999%
Support-response SLA	4-hour initial response, 8x5 (Business Standard)	24/7 SRE, plus the uptime guarantee
Allowed downtime / year	Customer-determined	~31 seconds
RTO	Customer-owned	Sub-1 minute
RPO	Customer-owned	Zero byte
Who operates the runtime	The customer	Akka

NVIDIA guarantees how fast it answers a support ticket; Akka guarantees that the system stays up and loses no data, and backs that with contractual indemnities.

Durable State the Runtime Holds

The NVIDIA stack serves models statelessly; durable agent state — conversation memory, in-flight task state, long-running plans — is not provided as managed infrastructure. The customer selects, wires, and operates a state store and a vector database, and owns their failure modes and latency.

Akka holds durable agent state in-memory at **4ms reads / sub-10ms writes**, sharded and replayable from its event journal, with no external store to provision. Akka is explicitly not a vector database or semantic knowledge layer; it is the durable execution substrate that holds the agent's working state across failures.

Cost

AI systems built with Akka are up to **90% cheaper to operate** than the equivalent Python-based stack — a function of the infrastructure required to run the same agentic transaction volume, not list price. The drivers are actor concurrency (~10T tokens/core/year vs ~2T; ~80% less compute than Python frameworks; Manulife: up to 300% more concurrency, 30–50% faster processing after porting from Python), shared compute, and micro-checkpointing.

NVIDIA AI Enterprise lists at **$4,500 per GPU per year** (or per-GPU-per-hour cloud) for the serving layer. That is the price of the inference-and-microservice bundle; the application layer above it — agent runtime, durable memory, streaming, APIs, observability, and governance — is provisioned and operated separately. Akka delivers that layer on one shared-compute runtime at a fixed annual fee.

Governance and the EU AI Act

NeMo Guardrails provides programmable content and topic rails — keeping a model on-topic, filtering unsafe outputs. That is content moderation, not governance enforcement. The NVIDIA stack publishes no inline runtime policy enforcement, no decision explainability, no human pause/override of a running process witnessed as it happens, no immutable interaction ledger, no pre-deployment classification, and no sealed audit artifact.

The penalties are enforceable now

Violation	Maximum Fine
Prohibited AI practices (Art. 5)	EUR 35M or 7% global turnover
High-risk obligations (Art. 9-15)	EUR 15M or 3% global turnover

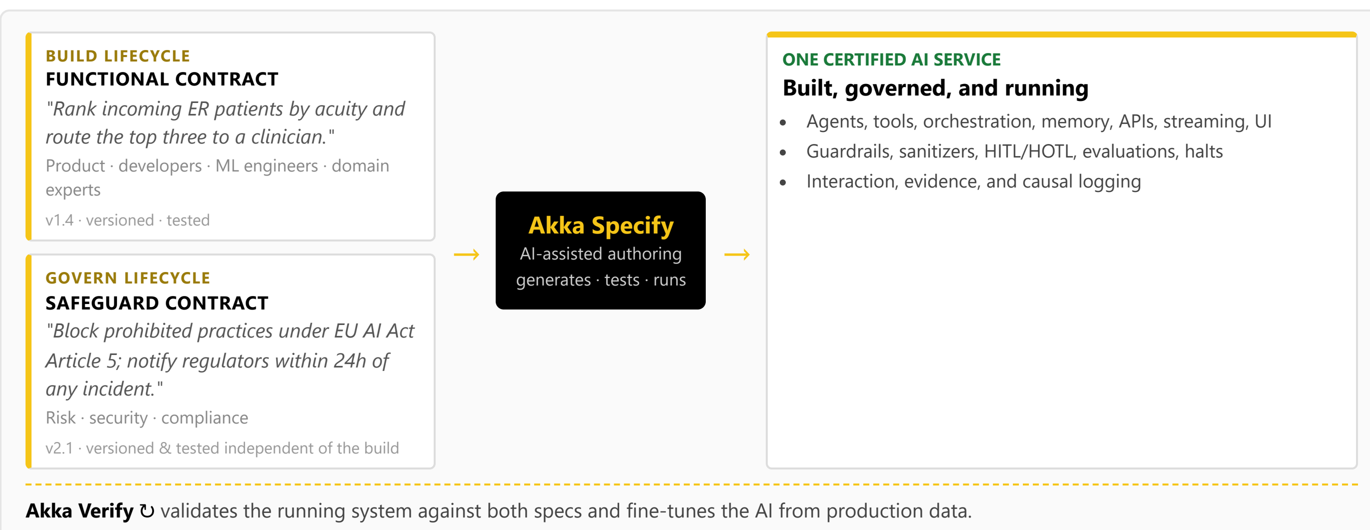
High-risk AI carries a 10-year logging-retention obligation (Art. 72).

How Akka governs

At the runtime: inline guardrails, policies, LLMs-as-a-judge, and sanitizers; hash-chained immutable evidence; HITL/HOTL control; classification against 190 regulations and 1,040 controls (671 carrying a financial penalty) before a system ships; multi-persona sign-offs; a sealed Governance Posture Package; and Akka Verify proving conformance from the running system. Governance the NVIDIA stack leaves to the customer, Akka enforces inline.

Two Lifecycles, One Certified System

Building on the NVIDIA stack means engineers wiring NIM endpoints, NeMo jobs, and toolkit code; there is no path for a product manager, domain expert, or risk officer to contribute, and no built-in governance lifecycle. Akka runs two independent lifecycles on one platform via **Akka Specify**:



The build lifecycle and the governance lifecycle are versioned and tested independently, by different audiences — an audience and a workflow the NVIDIA stack has no equivalent for.

Real-Time Streaming at Petabyte Scale

The NVIDIA stack has no application streaming engine; real-time agent feedback loops and high-throughput pipelines are provisioned separately. Akka's streaming is built into the runtime — continuous, backpressured, **petabyte-scale, in-memory**, with no external broker — powering both agent feedback loops and high-throughput data processing (the engine behind Tubi's real-time hyper-personalization at 5 billion tokens per second).

For the Buyer: Maturity, Hardware, and Accountability

Buyer concern	NVIDIA stack	Akka
Agent-platform maturity	NeMo Agent toolkit first released March 2025 (renamed from AIQ / Agent Intelligence toolkit), ~15 months old; a framework-agnostic glue layer over LangChain, LlamaIndex, and CrewAI — code-level instrumentation, not a hosted runtime	A production runtime: since 2007, 100,000+ deployments, 52 banks; 2 billion+ people touch an Akka-powered app daily
Hardware / cloud lock-in	Optimized for and dependent on NVIDIA GPUs and CUDA	Runs on any hardware, any cloud, on-prem, or sovereign cloud; portable specs, no GPU/CUDA dependency for the application layer
Scope of accountability	The serving layer; you integrate and operate the application, and own its uptime	One platform, one SLA, 24/7 SRE — Akka owns the running system
Risk transfer	Software support entitlements; availability is customer-owned	Availability and data-integrity guarantees backed by contractual indemnities
Certifications & audits	AI Enterprise production-branch support, monthly CVE patching, security notifications, SBOM/VEX	19 standards — SOC 2 II + public SOC 3, ISO 27001/42001, HIPAA, PCI DSS, GDPR, NIS2, DORA, EU AI Act, NIST AI RMF — plus annual pen tests, SBOMs, 40+ policies (trust.akka.io)
Budget predictability	Per-GPU licensing plus separately provisioned application infrastructure	Fixed annual fee finance can forecast

The NVIDIA agent-toolkit story is recent and code-level; its product maturity as an *application platform* is early, even though NVIDIA's serving products are mature and dominant. The decision is altitude and accountability: NVIDIA gives you the best serving layer to build an application around; Akka gives you the governed application platform that runs above it.

Akka Complements NVIDIA Inference

This is not a replacement story for inference. NVIDIA is the leading model-serving and acceleration layer, and Akka is not model serving, inference, RL, or a GPU runtime. An Akka agentic application can call NIM endpoints, use NeMo-customized models, and run on NVIDIA GPUs for the model tier — while Akka provides the governed application layer above it: native agents, durable state, streaming, APIs, the availability SLA, and runtime governance. The two fit together; they do not compete.

Customers Running Agentic and Real-Time Systems on Akka

Manulife
2,000
developers across 100 projects on one governed platform

Tubi
5B tok/s
real-time hyper-personalization engine

Swiggy
71ms
order-assignment AI, ~50% latency reduction

John Deere
1,000+
tractor sensors turned into real-time insight

Verizon
750%
order-processing capacity gain; 6s → 2.4s response

Common Questions

We already run NVIDIA NIM and AI Enterprise. Why add Akka?

NIM and AI Enterprise are an excellent serving layer, and Akka runs above them. NIM serves the model; Akka runs the agentic application that uses it — native agents, durable memory, streaming, APIs, an operated 99.9999% availability SLA, and runtime governance. Akka calls your NIM endpoints rather than replacing them.

Isn't the NeMo Agent toolkit NVIDIA's agent platform?

The NeMo Agent toolkit is an open-source library, first released in March 2025 (renamed from the Agent Intelligence toolkit / AIQ), for connecting and profiling teams of agents across frameworks like

Does NVIDIA AI Enterprise give us an uptime SLA?

AI Enterprise includes a support-response SLA — a 4-hour initial response, 8x5, plus security patches and maintenance branches. It does not include an availability or uptime SLA, because you run the software in your own infrastructure; the running system's uptime, RTO, and RPO are owned by the customer. Akka operates the runtime and guarantees 99.9999% availability with sub-1-minute RTO and zero-byte RPO.

Can we add governance on top of the NVIDIA stack?

NeMo Guardrails gives you content and topic rails. The EU AI Act expects more: enforcement inline to the runtime, immutable records

LangChain, LlamaIndex, and CrewAI. It is code-level instrumentation you host and operate, not a hosted runtime with an availability SLA, durable state, HA/DR, or governance enforcement. Akka is the governed runtime; the toolkit can sit inside an application Akka runs.

witnessed as they happen, human override on running processes, pre-deployment classification, and a sealed audit artifact. Akka embeds all of this — classification against 190 regulations and 1,040 controls before a system ships — rather than leaving it to the customer to assemble.

Sources

NVIDIA NIM: nvidia.com/en-us/ai-data-science/products/nim-microservices; developer.nvidia.com/nim — containerized GPU-accelerated inference microservices (TensorRT-LLM / vLLM / SGLang; OpenAI-compatible API); part of AI Enterprise; CUDA compute capability > 7.0 required

NVIDIA NeMo: nvidia.com/en-us/ai-data-science/products/nemo; developer.nvidia.com/blog/simplify-custom-generative-ai-development-with-nvidia-nemo-microservices — open-source generative-AI framework; microservices: Customizer, Evaluator, Guardrails, Retriever, Curator (built on CUDA-X)

NeMo Agent toolkit: github.com/NVIDIA/NeMo-Agent-Toolkit; developer.nvidia.com/nemo-agent-toolkit — open-source library, Apache-2.0 ("AS IS, WITHOUT WARRANTIES"); framework-agnostic (LangChain, LlamaIndex, CrewAI, Semantic Kernel, Google ADK); first release v1.0.0 March 2025

NeMo Agent toolkit history: forums.developer.nvidia.com/t/335881; infoworld.com/article/3851326 — renamed from Agent Intelligence toolkit / AIQ; launched March/April 2025

NVIDIA Blueprints: blogs.nvidia.com/blog/nim-agent-blueprints; nvidianews.nvidia.com (NIM Agent Blueprints launch) — reference AI workflows: sample apps + reference code + Helm charts; deployed in production via AI Enterprise

NVIDIA AI Enterprise pricing & support: docs.nvidia.com/ai-enterprise/planning-resource/licensing-guide/latest/pricing.html; dell.com SKU ac566091 ($4,500/GPU/year, includes Standard 8x5 support; also per-GPU-per-hour cloud); nvidia.com/en-us/support/enterprise (Business Standard: 4-business-hour Sev-1 response) — support-response SLA, not an uptime SLA; 99% target applies only to hosted DGX Cloud

NVIDIA AI Enterprise lifecycle/security: docs.nvidia.com/ai-enterprise/lifecycle/latest/choosing-a-branch.html — production branch, monthly CVE patches, SBOM/VEX, container/model signing

Akka platform, governance, trust & performance: per akka-facts.md — trust.akka.io; akka.io/blog/go-slow-to-go-fast; 99.9999% availability, sub-1-min RTO, zero-byte RPO (contractual indemnities); 4ms / sub-10ms state; 190 regulations / 1,040 controls / 671 with financial penalty; 100,000+ deployments / since 2007; profitable; Dell Technologies Capital

NVIDIA claims are drawn from NVIDIA's own documentation, blog, and newsroom, and NVIDIA AI Enterprise licensing material. Akka claims reflect Akka's published capabilities. NVIDIA is the leading model-serving and inference layer; Akka operates as the governed application platform above it and does not provide inference or model serving. · June 2026